A U-ADMISSIBLE BLOWUP AND A SURJECTIVE OPEN ON WHICH THE STRICT TRANSFORM OF A FINITE TYPE MODULE BECOMES FINITE LOCALLY FREE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; no macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0811 (sources/stacks-flat/pinned-original-source.tex line 8782).

Allowed standard prerequisites (editorial).

- The successive-blowup reductions and separation of disjoint opens by an admissible blowup. Same chapter, Remark labelled remark-successive-blowups (source line 8611) invoked at proof lines 8810, 8837 and 8839, and Divisors, Lemma labelled divisors-lemma-separate-disjoint-opens-by-blowing-up (Stacks tag 080P) at proof lines 8826-8827. Allow reductions of the problem under the two properties that are invariant under strict transform and local on S, and provide the blowup that separates the level sets of the rank function.
- Flatness at the generic points of the fibres is preserved by strict transform, and locally the complement of U is cut out by a nonzerodivisor. Divisors, Lemma labelled divisors-lemma-strict-transform-flat (Stacks tag 080F) with Morphisms, Lemma labelled morphisms-lemma-base-change-module-flat (Stacks tag 01U8) at proof lines 8806-8808, and Divisors, Lemma labelled divisors-lemma-characterize-effective-Cartier-divisor (Stacks tag 01WS) at proof lines 8840-8841. Supply the invariance of the hypothesis and the reduction to S affine with the complement of U defined by one nonzerodivisor.
- Flatness and freeness at a point of the fibre, and the conclusion of the reduced case. More on Morphisms, Lemma labelled more-morphisms-lemma-flat-and-free-at-point-fibre (Stacks tag 080Q) at proof lines 8820-8821, and Same chapter, Lemma labelled lemma-helper-blowup-affine-space (source line 8668) at proof lines 8844-8845. Supply the local constancy of the rank function and the final affine computation; the construction of V and the reduction to this case are inside the counted span.

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Source reference. Original lemma 080F: sources/stacks-chow/tags.txt, source lines 10180–10180. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8807 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01U8: sources/stacks-chow/tags.txt, source lines 2401–2401. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8808 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original remark 080Y: sources/stacks-flat/pinned-original-source.tex, source lines 8611–8611. Same-chapter pointer into the author environment remark at source line 8611; invoked at proof line 8810 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 080Q: sources/stacks-chow/tags.txt, source lines 10190–10190. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8821 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 080P: sources/stacks-chow/tags.txt, source lines 10189–10189. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8827 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01WS: sources/stacks-chow/tags.txt, source lines 2493–2493. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8841 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0810: sources/stacks-flat/pinned-original-source.tex, source lines 8668–8668. Same-chapter pointer into the author environment lemma at source line 8668; invoked at proof line 8845 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let S be a quasi-compact and quasi-separated scheme. Let $X \to S$ be a morphism of schemes. Let $\mathcal{F}$ be a quasi-coherent module on X. Let $U \subset S$ be a quasi-compact open. Assume*

(1) *$X \to S$ is affine, of finite presentation, flat, geometrically integral fibres,*
(2) *$\mathcal{F}$ is a module of finite type,*
(3) *$\mathcal{F}_U$ is of finite presentation,*
(4) *$\mathcal{F}$ is flat over S at all generic points of fibres lying over points of U.*

Then there exists a U-admissible blowup $S' \to S$ and an open subscheme $V \subset X_{S'}$ such that (a) the strict transform $\mathcal{F}'$ of $\mathcal{F}$ restricts to a finitely locally free $\mathcal{O}_V$-module and (b) $V \to S'$ is surjective.

Proof. Given $\mathcal{F}/X/S$ and $U \subset S$ with hypotheses as in the lemma, denote P the property "$\mathcal{F}$ is flat over S at all generic points of fibres". It is clear that P is preserved under strict transform, see Divisors, Lemma 080F and Morphisms, Lemma 01U8. It is also clear that P is local on S. Hence any and all observations of Remark 080Y apply to the problem posed by the lemma.

Consider the function $r : U \to \mathbf{Z}_{\geq 0}$ which assigns to $u \in U$ the integer

$$r(u) = \dim_{\kappa(\xi_u)}(\mathcal{F}_{\xi_u} \otimes \kappa(\xi_u))$$

where ξ_u is the generic point of the fibre X_u. By More on Morphisms, Lemma 080Q and the fact that the image of an open in X_S in S is open, we see that $r(u)$ is locally constant. Accordingly $U = U_0 \amalg U_1 \amalg \ldots \amalg U_c$ is a finite disjoint union of open and closed subschemes where r is constant with value i on U_i. By Divisors, Lemma 080P we can find a U-admissible blowup to decompose S into the disjoint union of two schemes, the first containing U_0 and the second $U_1 \cup \ldots \cup U_c$. Repeating this $c - 1$ more times we may assume that S is a disjoint union $S = S_0 \amalg S_1 \amalg \ldots \amalg S_c$ with $U_i \subset S_i$. Thus we may assume the function r defined above is constant, say with value r.

By Remark 080Y we see that we may assume that we have an effective Cartier divisor $D \subset S$ whose support is $S \setminus U$. Another application of Remark 080Y combined with Divisors, Lemma 01WS tells us we may assume that $S = \mathrm{Spec}(R)$ and $D = \mathrm{Spec}(R/(f))$ for some nonzerodivisor $f \in R$. This case is handled by Lemma 0810. $\square$